



Semester DB Project: Phase I

1. Thoroughly document the specific revisions and improvements to be made to your group's project. Include explanations for each change, detailing why it was made and how it enhances the project.
 - Based on the feedback we received for Phase 1, we plan to make several changes to reflect how research work, tasks, roles, and people are organized in our database. As of right now, role is stored as an attribute of the person entity, but roles can change depending on context (Someone may be in more than one research group, where the same person can have different roles). To address this issue, we plan to remove role from person and instead keep it within relationship sets like person_group and project_member, where role will be more accurately depicted based on context. The task entity currently includes the theory attribute, which peer review proved to be confusing. Since tasks are based around actions, we plan to either rename this attribute with a more transparent name, or move the attribute entirely to either the experiment or literature entities. Based on our peer reviewed feedback, we also plan to better explain how users will interact with our database in order to track tasks, progress, collaboration, and search for data, including examples of how users can add information and make changes to their work.
2. - Identify weaknesses or concerns raised in the feedback and explain how your group will address them. Prioritize revisions that will have the most significant impact on improving the overall quality and functionality of your project.
 - One of the major feedback we will prioritize is the user interface. I feel this revision will be the most important as it will enable users of our database to easily access and manipulate data. One weakness of our project that was raised in the feedback we received was our attributes we created in the diagram, specifically the theory attribute in the task entity. Another point of weakness for our project seems to be our relationship sets and how we construct them. To address this our group should specify and go over the type of relationships each entity can have with each other and how these relationships can interact.
3. Updated entity sets and their attributes with primary keys underlined.

Entity	Primary Key	Attributes
Department	<u>department_id</u>	name, abbreviation
Person	<u>person_id</u>	name, email
Research_Group	<u>group_id</u>	name, focus
Project	<u>project_id</u>	title, description, start_date, end_date, status
Experiment	<u>experiment_id</u>	title, objective, description, start_date, end_date, status
Task	<u>task_id</u>	title, description, status, due_date, completed_date, progress
Literature	<u>lit_id</u>	title, authors, year, journal, doi, url, theory

Notes:

Person: role removed ; now tracked in person_group and project_member relationships.

Task: theory removed

Literature: theory moved here to store research content.

Experiment: objective added to clarify purpose of experiment.

4. Updated relationships and constraints.

We removed **leads** because leadership is already captured by **person_group.role**, and keeping both would duplicate the same information and make the model harder to interpret.

Relationship	Entities Connected	Type	Relationship Attributes
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person_dept	Person ↔ Department	M:1	—
research_dept	Research_Group ↔ Department	M:1	—
person_group	Person ↔ Research_Group	M:N	role, join_date
group_project	Research_Group ↔ Project	1:M	—
project_member	Person ↔ Project	M:N	role, join_date
project_tasks	Project ↔ Task	1:M	—
task_assignment	Person ↔ Task	M:N	assigned_date
project_expt	Project ↔ Experiment	1:M	—
project_literature	Project ↔ Literature	M:N	—
expt_literature	Experiment ↔ Literature	M:N	—

5. Updated E-R Diagram reflecting the changes made to the project structure.
6. Descriptions of the sources from which you intend to collect data for your database, possible high-level data modification operations, and the types of questions your stored data will be able to answer.

- a. Sofia's source material (paired down for ease)
- b. Theoretical research projects as proof of concept
- c. Operations:
 - i. INSERT: users can create tasks, add literature
 - ii. UPDATE: users can change due dates, status, progress
 - iii. DELETION: Remove completed/null tasks
 - iv. Special cases: instructors can modify at the project level, reassign tasks, modify users
- d. Example Questions:
 - i. 'Can I see all articles published by Angelo Ricarte in 2025'
 - ii. 'Has Yuvraj completed task 12'
 - iii. 'What tasks are assigned to me'
 - iv. 'What tasks has no one assigned'
 - v. 'Who is the PI for the Gravitational Waves project'